Lecture 12

2/16/26

Homework Issues

Start lecture w/ quick discussion of two homework problems

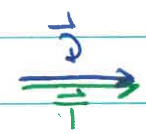
5.4 DF μscopy - Vector Diagram

The general relationship among $\vec{T}$, $\vec{Z}$, and $\vec{B}$ is given by the equation

$$\vec{T} = \vec{Z} + \vec{D}$$

In DF μscopy, $\vec{Z} = 0$, by design, so $\vec{T} = \vec{D}$

The associated vector diagram has both vectors superimposed

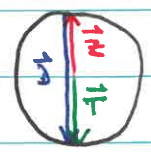


5.5 PC μscopy - Vector Diagram

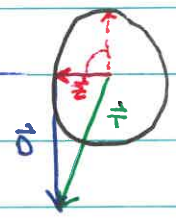
What is the magnitude of $\vec{T}$ if the specimen induces a 180° (π rad) phase shift in the $\vec{D}$ wave?

Draw vector diagram & compute $\vec{T}$

BF μscopy



Positive PC μscopy



Left (BF) diagram shows $|\vec{D}| = 2|\vec{Z}|$

Right (+PC) diagram shows (using Pythagorean Theorem)

$$|\vec{T}|^2 = |\vec{Z}|^2 + |\vec{D}|^2 \xRightarrow[\vec{D} = 2\vec{Z} \text{ \& solve}]{\text{Substitute}} |\vec{T}|^2 = |\vec{Z}|^2 + 4|\vec{Z}|^2$$

$$= 5\,|\vec{Z}|^2$$

$$\boxed{|\vec{T}| = \sqrt{5}\,|\vec{Z}|}$$

Review Theoretical Foundations of Phase Contrast

The way to understand phase contrast, a Nobel Prize winning technique, is to implement Abbe's viewpoint

$$\boxed{\vec{T} = \vec{Z} + \vec{D}}$$

Expression applies at each resolvable point in image

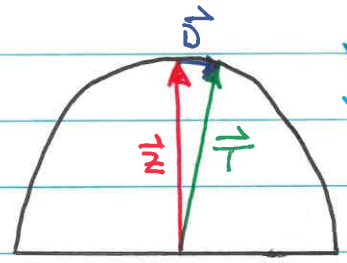
$\vec{D}$ varies across specimen (e.g., cell) because it represents the light diffracted or otherwise perturbed by the sample

PC uses interference to create contrast

PC μscopy uses a trick (advancing or retarding the $\vec{Z}$ wave) to cause the $\vec{D}$ wave and the $\vec{Z}$ wave to inferere to create a $\vec{T}$ wave that reflects specimen variations

ACTIVITY 5

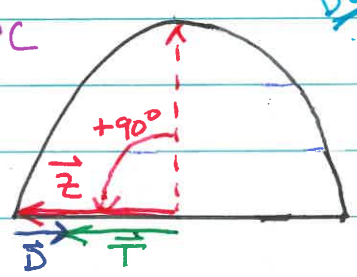
BF



No interference

- $|\vec{T}| = |\vec{Z}|$
- Because there's no absorption $\Rightarrow$ No contrast

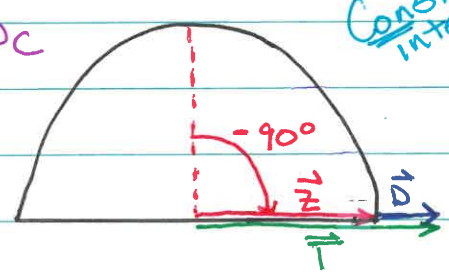
Pos PC



Destructive interference

- Advance $\vec{Z}$ by 90° counterclockwise
- Do NOT alter $\vec{D}$
- $|\vec{T}| < |\vec{Z}|$
- Cell darker than background (where there's no cell) because $|\vec{T}|$ smaller than $|\vec{Z}|$

Neg PC



Constructive interference

- Retard $\vec{Z}$ by 90° clockwise
- Do NOT alter $\vec{D}$
- $|\vec{T}| > |\vec{Z}|$
- Cell brighter than background (where there's no cell) because $|\vec{T}|$ larger than $|\vec{Z}|$

The $\vec{E}$ wave can be advanced or retarded with tricks in the FFP of the condenser & the BFP of the objective

Experimental "Trick"

(1) Make $\vec{E}$ annular in FFP of condenser using a mask w/ an annular (ringlike) opening

(2) Make an annular etch in an otherwise uniform glass plate positioned in the BFP of the objective because $\vec{E}$ will also be annular in this plane. Etch is about 250 nm deep.

(3) $\vec{D}$ is spread out all over BFP of objective so $\vec{D}$ is mostly unshifted. A little diffracted light will pass thru the etch & get phase shifted, giving rise to a "halo artifact"

Polarization Microscopy

Defer (or omit) discussion of polarization μscopy in favor of starting on fluorescence microscopy now

Fluorescence Microscopy

Fluorescence μscopy is currently about 80% of life sciences μscopy. It's also moving into medicine (e.g., as discussed in last weeks Biochemistry Seminar w/ Trace Biosciences)

Why is Fluorescence so special?

(1) Sensitive. Signal viewed against dark background.

Compared phase contrast & fluorescence images of the same nerve growth cone (Power Point slides & Fig. 6.1 in text)

(2) Contrast. Bright against dark, as mentioned under (1), but also color vs color. The human eye is supersensitive to color variation.

(3) Living specimens. Fluorescence is compatible with studying living specimens

(4) Selectivity. Fluorescence may be used selectively to look at specific cellular components by labeling each w/ a different color.

Most cellular components are NOT intrinsically fluorescent, so must label components of interest w/ a fluorescent molecule ("fluorophore")

Example 1 Fluorescent growth cone (see PowerPoints)

Red - Actin
Blue - Microtubules
Green - Dense core granules

Example 2 Violet - Chromosomes (labeled w/ DAPI)

What are the disadvantages of fluorescence?

(1) Need to label. Samples must be labeled to study w/ fluorescence. This can be difficult, time-consuming, etc. Samples do NOT require labeling for BF, DF, PC μscopy.

(2) Need intense illumination. Fluorescent samples must be excited to induce fluorescence. The bright excitation light can cause various problems.

(A) Photodamage - light can kill sample (cells)

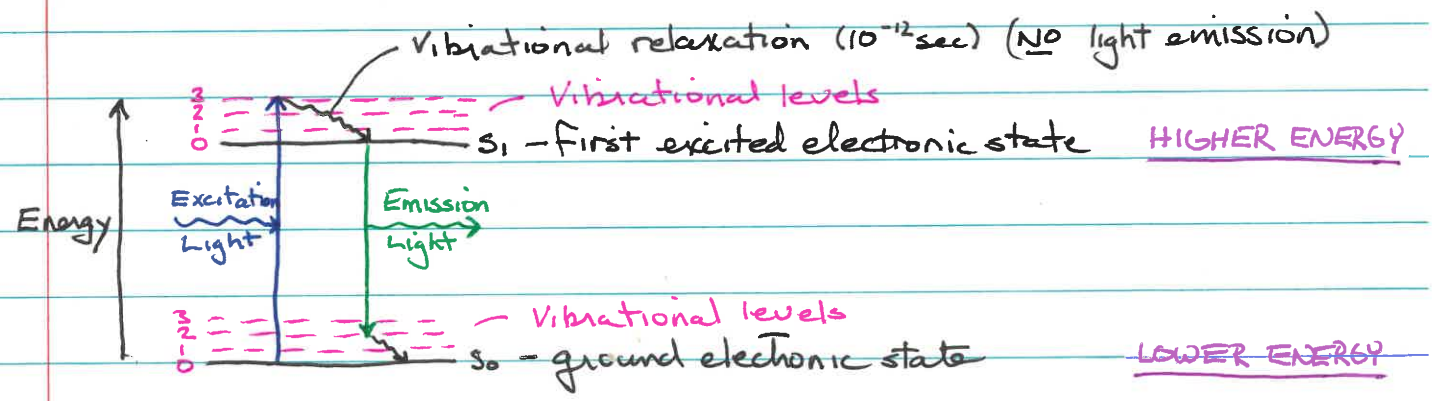
(B) Photobleaching - light can destroy fluorophores, leading to loss of signal

Photobleaching can be caused by reactive oxygen & reduced or prevented by deoxygenating samples.

Physics of Fluorescence

{ Very important
Will test on subject

Jablonski Diagram - diagram of energy levels showing transitions that lead to the absorption & emission of light



Fluorophore = molecule that fluoresces

Molecules usually sit in their lowest energy state

Negatively charged electrons are attracted to positively charged protons in nucleus

The electronic energy levels are correlated with how far the electrons are from the nucleus

Ground state - electrons closer to nucleus — Lower energy
Excited state(s) - electrons farther from nucleus — Higher energy

There are other sources of energy available to molecules

Energy Sources
- Electronic
- Vibration
- Rotation

} Important in fluorescence

} NOT important in fluorescence

Vibrational energy levels are lower than electronic energy levels

In fluorescence molecule starts in lowest energy level (electronic + vibrational) & is excited into an excited energy level (electronic and sometimes vibrational) with light energy

The fluorophore usually goes into a higher-energy vibrational level. This energy is lost super quickly (10^{-12} sec) to heat, leaving fluorophore in lowest vibrational level of excited electronic state

The fluorophore will fall back into the ground electronic state after about 10^{-9} sec

The "lost" energy, for a good fluorophore, goes into an emitted photon. Do <u>NOT</u> want the energy to go into heat (although that often happens)

The energy of the emitted fluorescence photon will be lower than the energy of the excitation photon (due in part to energy lost during "vibrational relaxation" between excitation & emission)

$\boxed{\text{Energy shift} \Rightarrow \text{Color shift}}$

$\Rightarrow$ Excitation & emission photons have different colors

Physics 390 In-class Activity 5

Brightfield and Phase Contrast

Name: _____

BF Vector Diagram	Positive PC Vector Diagram	Negative PC Vector Diagram
Contrast Analysis*	Contrast Analysis*	Contrast Analysis*

* Please describe (1) the grayscale variation within the specimen, and (2) the grayscale level of the specimen relative to background. Justify your statements using the vector diagram!